Stats 544, Homework 6

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Exercise 7-1.

It is well-known that hypertension (high blood pressure) and type II diabetes are major risk factors for cardiovascular diseases. Suppose in a group of patients with both hypertension and diabetes, a bivariate normal distribution is used to model the diastolic blood pressure (B) and fasting plasma glucose (G) with the following statistic: $E(B)$=80 mmHg, $SD(B)$=7.5 mmHg, $E(G)$=117 mg/dL, and $SD(G)$=24 mg/dL.

Also assume that the correlation between the diastolic blood pressure and fasting plasma glucose is 0.3.

A. Estimate the proportion of patients whose diastolic blood pressure at least 80 AND the fasting glucose is at most 126.

Solution: The calculation we want is P(B>= 80, G <= 126)

```
# Run once if needed:
# install.packages("mvtnorm")

library(mvtnorm)
```

```
Warning: package 'mvtnorm' was built under R version 4.5.2
```

```
mu <- c(80, 117)

Sigma <- matrix(
  c(7.5^2, 0.3 * 7.5 * 24, 0.3 * 7.5 * 24, 24^2),
  nrow = 2,
  byrow = TRUE
)

p_A <- as.numeric(
  pmvnorm(
    lower = c(80, -Inf),
    upper = c(Inf, 126),
    mean = mu,
    sigma = Sigma
  )
)

p_A
```

```
[1] 0.2779861
```

```
# Approximately 0.2779861
```

B. Suppose your colleague ignored the correlation and treated the diabolic blood pressure and the fasting glucose as two independent variables. Then they estimated the proportion of patients whose diastolic blood pressure at least 80 AND the fasting glucose is at most 126. What would be their estimate ? How would you explain the discrepancy between your and their answers?

Solution: Under the independence assumption, the joint probability is the product of the marginal probabilities: P(B>=80, G<=126) = P(B>=80)P(G<=126), and since 80 is the mean, P(B<=80)=0.5

Why the answer differs from the part A where the variables are treated as correlated: The variables have a positive correlation meaning that higher blood pressure tends to occur with higher fasting glucose. For patients with blood pressure at least 80, gulcose is less likely to remain below 126 than it is in the overall population. As a result the assumption overestimates the requested proportion by about 4.51%.

```
p_B <- pnorm(80, mean = 80, sd = 7.5, lower.tail = FALSE) *
       pnorm(126, mean = 117, sd = 24)

p_B
```

```
[1] 0.3230849
```

```
# 0.3230849

# Difference in percentage points
100 * (p_B - p_A)
```

```
[1] 4.509883
```

```
# Approximately 4.509883
```

Exercise 8-1.

Suppose the lifetime of a certain type of light bulb can be modeled by a gamma distribution with shape parameter α=2 and rate parameter λ=1/500.

A. Find the (theoretical) mean and standard deviation of the distribution. B. Generate 1000 lifetimes and report the summary statistics. C. Select lifetimes that are at least 150 hours and reduce that amount from each remaining observation. Compare the summary statistics with that in (b). D. Overlay the histograms using data in (b) and (c). Do they match well? Explain.

```
# A. the mean calculated by the following: E[X] = α/λ;
# and the SD by Var(X)^0.5 = sqrt(α/λ^2) = sqrt(α) /λ
```

```
alpha <- 2
lambda <- 1/500

a_mean <- alpha / lambda
a_sd <- sqrt(alpha/(lambda * lambda))

a_mean # 1000
```

```
[1] 1000
```

```
a_sd # 707.1068
```

```
[1] 707.1068
```

```
# B. Generate 1000 lifetimes and report the summary statistics
# Set a seed for reproducibility
set.seed(42)

# Generate 1000 random values with shape = alpha and scale = 1/lambda
# Note: R accepts a scale parameter, but scale is the reciprocal of rate,
# so scale = 500 not 1/500
simulated_data <- rgamma(1000, shape = alpha, scale = 1/lambda)
summary(simulated_data)
```

```
   Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
  6.295  452.561  820.899  965.146 1321.466 4873.299 
```

```
#     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
#   6.295  452.561  820.899  965.146 1321.466 4873.299 
```

```
# C. Keep lifetimes >= 150 hours, then subtract 150 hours
# The expression uses logical indexing to select the observations
# that satisfy the condition.
# We are calculating the remaining bulbs that have already lasted 150 hours,
remaining_lifetimes <- simulated_data[simulated_data >= 150] - 150

summary(remaining_lifetimes)
```

```
    Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
  0.5297  353.9119  711.7454  854.9757 1202.1398 4723.2993 
```

```
#     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
#   0.5297  353.9119  711.7454  854.9757 1202.1398 4723.2993 
```

The expression above uses logical indexing to select the observations that satisfy the condition to calculate the remaining bulbs that have already lasted 150 hours.

```r
# Compare the summary statistics
comparison <- rbind(
  Original = c(
    N = length(simulated_data),
    summary(simulated_data),
    SD = sd(simulated_data)
  ),
  Remaining = c(
    N = length(remaining_lifetimes),
    summary(remaining_lifetimes),
    SD = sd(remaining_lifetimes)
  )
)

round(comparison, 3)
```

```
             N  Min. 1st Qu.  Median    Mean  3rd Qu.     Max.      SD
Original  1000 6.295 452.561 820.899 965.146 1321.466 4873.299 668.596
Remaining  956 0.530 353.912 711.745 854.976 1202.140 4723.299 656.861
```

D. Overlay the histograms using data in (b) and (c).

```r
# Plot the histogram of the simulated data

h_b <- hist(simulated_data, breaks = 30, plot = FALSE)
h_c <- hist(remaining_lifetimes, breaks = 30, plot = FALSE)

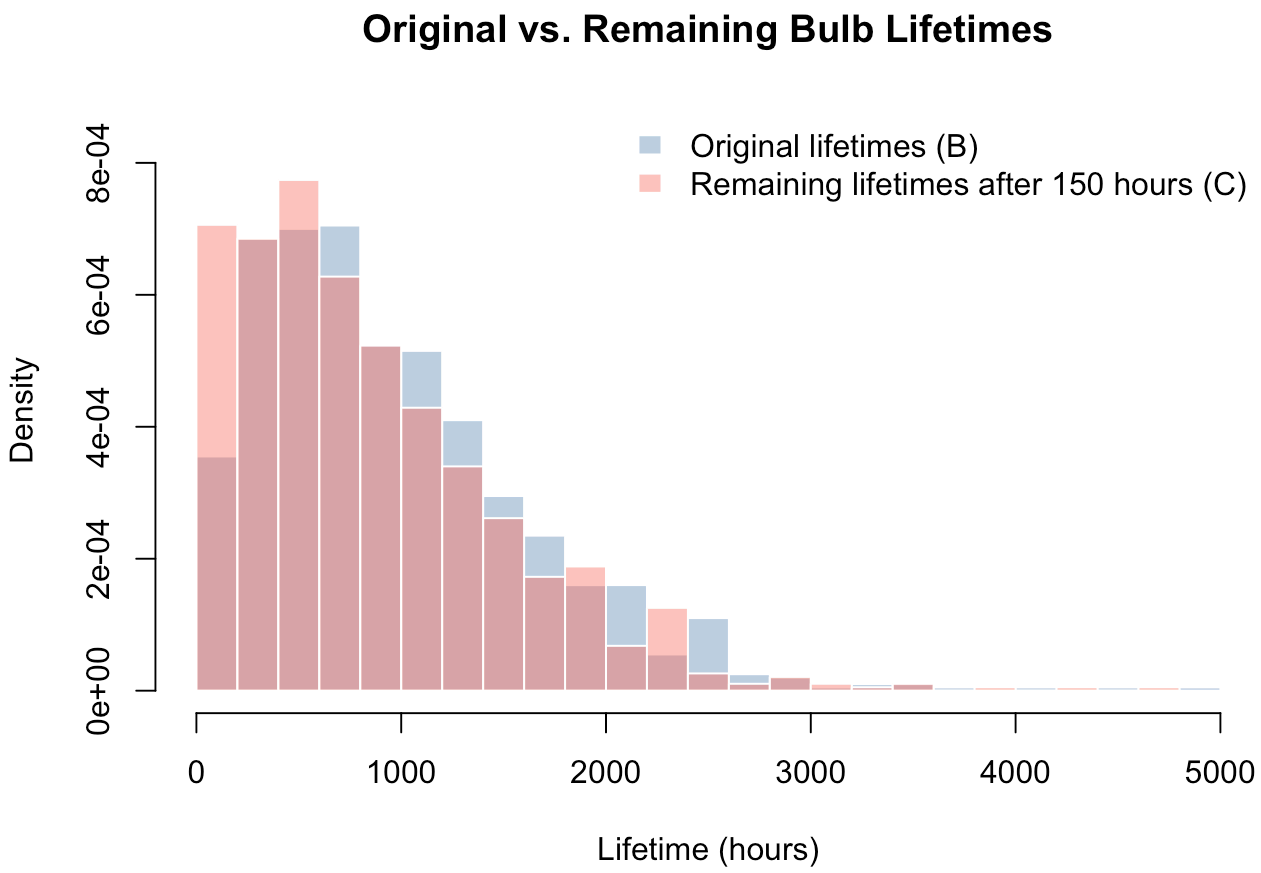
y_max <- max(c(h_b$density, h_c$density))


# Plot B
plot(h_b,
     freq = FALSE,
     col = adjustcolor("steelblue", alpha.f = 0.4),
     border = "white",
     ylim = c(0, y_max * 1.1),
     main = "Original vs. Remaining Bulb Lifetimes",
     xlab = "Lifetime (hours)",
     ylab = "Density")

# Overlay C
plot(h_c,
     freq = FALSE,
     col = adjustcolor("tomato", alpha.f = 0.4),
     border = "white",
     add = TRUE)

legend("topright",
       legend = c("Original lifetimes (B)",
                  "Remaining lifetimes after 150 hours (C)"),
```

```
        fill = c(adjustcolor("steelblue", alpha.f = 0.4),
                 adjustcolor("tomato", alpha.f = 0.4)),
        border = "white",
        bty = "n")
```



Explanation: First, by removing lifetimes below 150, we eliminated the smallest observations which raised the mean of the retained original lifetimes. Second, by subtracting 150 shifts every observation down by 150, therefore, the new mean is not simply the original mean of 965.146 minus 150.

Subtracting a constant does not change the standard deviation of the retained observations, but removing observations can cahnge it relative to the full sample. We can verify with:

```
sd(simulated_data[simulated_data >= 150])
```

```
[1] 656.8612
```

```
sd(remaining_lifetimes)
```

```
[1] 656.8612
```

```
# These are equal, apart from possible floating-point rounding.
```

3.

Exercise, p.6, lecture slides: Order statistics and transform: Let X1, X2, X3, X4 be a random sample from an exponential distribution with mean 1. A. Determine the distribution function of X2. B. Evaluate P(X2 <= 0.25) C. Plot the density function of X2.